Ethan Miao

206-327-5868 | shiqi111@uw.edu | github.com/shiqi1125 | Linkedin

Education

University of Washington — Seattle, WA
M.S. in Electrical and Computer Engineering — *Sep 2026 – Expected Jun 2028*

Huazhong Agricultural University — Wuhan, China
B.S. in Computer Science and Technology; GPA: 3.94/4.00 — *Sep 2022 – Jun 2026*

Professional Experience

ZWSOFT — Wuhan, China
Mechanical R&D - Software Development Intern — *May 2026 – Aug 2026*

- Developed a **C++ AI Assistant plugin** for ZWCAD and designed an asynchronous IPC layer between CAD and an independent AI service for CAD drawing processing and knowledge retrieval; added structured logging and end-to-end Request ID tracing for cross-process observability, troubleshooting, and debugging.
- Built a **FastAPI** gateway for a distributed backend spanning retrieval, inference, knowledge-base, and internal services; implemented **REST/gRPC** routing and service-to-service communication, using PostgreSQL for persistent task state and Redis for caching, reducing retrieval P95 latency by **54%** from 460 ms to 210 ms.
- Engineered a high-concurrency pipeline using asynchronous I/O, **RabbitMQ**, and worker pools to decouple online requests from long-running parsing and indexing jobs; added connection pooling, rate limiting, and timeouts, sustaining 500+ concurrent requests and improving throughput by **27%**.
- Containerized backend services with **Docker** and deployed them to an internal **Kubernetes** cluster with load balancing, health checks, automatic restarts, and horizontal scaling; improved operational excellence with Prometheus and Grafana, reducing mean recovery time from **10 to 6 minutes**.

Huazhong Agricultural University — Wuhan, China
Research Assistant – AI Algorithms — *Jan 2025 – Apr 2025*

- Fine-tuned a pretrained **plant genomic foundation model** using **LoRA** ($r = 16$) on attention layers and a regression head, applying cosine learning-rate decay while updating under **2%** of model weights.
- Fused DNA sequence embeddings with 3D structural features using a **Graph Attention Network** and gated attention, improving Pearson r from **0.71 to 0.77** and reducing prediction MAE by **11.8%**.
- Evaluated frozen, LoRA-tuned, and multimodal models on held-out chromosomes using RMSE and **residual error analysis**; refined gene-level expression predictions, reducing systematic prediction bias by **16.4%**.
- Deployed the fine-tuned model on **AWS SageMaker** with model versioning and CloudWatch monitoring; optimized model loading and dynamic batching to increase inference throughput by 2.1× at a fixed GPU.

Projects

Agentic AI Cloud Storage Platform

- Architected a multi-tenant cloud storage platform with **7 microservices** and **40+** REST APIs, integrating S3 multipart uploads, Redis distributed locks, and AWS ALB to support 20 GB files and **300+** RPS.
- Engineered a **hybrid RAG** pipeline combining dense vector retrieval, BM25, and cross-encoder reranking with **OpenSearch** across **1M+** records, improving Recall@5 from **72% to 86%** over dense-only retrieval.
- Developed a stateful **AI agent** using **LangGraph** with dynamic planning, tool calling, and workflow checkpointing, achieving **87% task success** on multi-step file discovery, cross-document reasoning, and version comparison.
- Exposed file operations as standardized **MCP tools** with permission-aware execution, human-in-the-loop approval, and automated recovery, achieving **98.5% tool-call success** and **92% recovery rate** in failure-injection tests.

Skills

Programming Languages: Python, C, C++, C#, Java, Rust, JavaScript, TypeScript, SQL
AI & Machine Learning: AI Agents, RAG, MCP, Vector Search, LangChain, LangGraph, PyTorch, Fine-tuning
Backend & Distributed Systems: FastAPI, REST APIs, gRPC, PostgreSQL, MySQL, Redis (NoSQL), RabbitMQ
Cloud Services & DevOps: AWS, Docker, Kubernetes, Nginx, GitHub Actions, Prometheus, Grafana, Linux, Git
Professional Skills: Written and Verbal Communication, Collaboration, Analytical Problem Solving

Publications

Wang S, Zhao S, Miao S, Kwashie S, Bewong M, Hu J, Nofong VM, Feng Z. FastER: On-demand Entity Resolution in Property Graphs. International Semantic Web Conference 2025 Oct 29 (pp. 177-195). Cham: Springer Nature Switzerland.